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Get Data From Text/CSV Recent Sources From Web Existing Connections Refresh All Workbook Links

Queries & Connections Properties Workbook Links

Sort Filter Clear Reapply Advanced

Flash Fill Remove Duplicates Consolidate Data Model What-If Analysis Forecast Subtotal Outline Analysis

H10

Customer ID	name
Id1	Hawks, Ms. Kelly
Id2	Lehner, Mr. Matthew D
Id3	Lu, Mr. Phil
Id4	Osborne, Ms. Kelsey
Id5	Kadala, Ms. Kristyn
Id6	Baker, Mr. Russell B.
Id7	Macpherson, Mr. Scott
Id8	Hallman, Mr. Stephen
Id9	Moran, Mr. Patrick R.
Id10	Benner, Ms. Brooke N.
Id11	Fierro Vargas, Ms. Paola Andrea
Id12	Franz, Mr. David
Id13	Foster, Mr. Wade
Id14	Tenorio, Mr. Franklin
Id15	Rios, Ms. Leilani M.
Id16	Viau-Dupuis, Mr. Philippe
Id17	Cronin, Ms. Jennifer A.
Id18	Noordstar, Ms. Christina M.
Id19	Boudalia, Mr. Said Sr.
Id20	Flor, Mr. John
Id21	Fennon, Mr. Myles
Id22	Hribar, Ms. Madelyn C.
Id23	Tassello, Ms. Nicole
Id24	Mauricette, Mr. Eric A.

Customer Names Medical Examinations Hospitalisation Details

Ready Accessibility: Good to go

- Checked for duplicate values in customer name sheet. No duplicate values were found in the dataset.

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B1

Customer ID	Mr.	Ms.	Last Name
Id1	Mr.	Ms.	Kelly
Id2	Mr.	Ms.	Matthew D
Id3	Mr.	Ms.	Phil
Id4	Mr.	Ms.	Kelsey
Id5	Mr.	Ms.	Kristyn
Id6	Mr.	Ms.	Russell B.
Id7	Mr.	Ms.	Scott
Id8	Mr.	Ms.	Stephen
Id9	Mr.	Ms.	Patrick R.
Id10	Mr.	Ms.	Brooke N.
Id11	Mr.	Ms.	Paola Andrea
Id12	Mr.	Ms.	David
Id13	Mr.	Ms.	Wade
Id14	Mr.	Ms.	Franklin
Id15	Mr.	Ms.	Leilani M.
Id16	Mr.	Ms.	Philippe
Id17	Mr.	Ms.	Jennifer A.
Id18	Mr.	Ms.	Christina M.
Id19	Mr.	Ms.	Said Sr.
Id20	Mr.	Ms.	John
Id21	Mr.	Ms.	Myles
Id22	Mr.	Ms.	Madelyn C.
Id23	Mr.	Ms.	Nicole
Id24	Mr.	Ms.	Eric A.

Customer Names Medical Examinations Hospitalisation Details

Ready Accessibility: Good to go

- Split the Name column into three separate columns.

Customer ID	Title	first name	last name
Id1	Ms	Kelly	Hawks
Id2	Mr	Matthew D	Lehner
Id3	Mr	Phil	Lu
Id4	Ms	Kelsey	Osborne
Id5	Ms	Kristyn	Kadala
Id6	Mr	Russell B	Baker
Id7	Mr	Scott	Macpherson
Id8	Mr	Stephen	Hallman
Id9	Mr	Patrick R	Moran
Id10	Ms	Brooke N	Benner
Id11	Ms	Paola Andrea	Fierro Vargas
Id12	Mr	David	Franz
Id13	Mr	Wade	Foster
Id14	Mr	Franklin	Tenorio
Id15	Ms	Lellani M	Rios
Id16	Mr	Philippe	Viau-Dupuis
Id17	Ms	Jennifer A	Cronin
Id18	Ms	Christina M	Noordstar
Id19	Mr	Said Sr	Boudalia
Id20	Mr	John	Flor
Id21	Mr	Myles	Fennon
Id22	Ms	Madelyn C	Hribar
Id23	Ms	Nicole	Tassello
Id24	Mr	Eric A	Mauricette

- Added appropriate headings for the three columns: Title, First Name, and Last Name. Then, added borders to the columns.

Customer ID	BMI	HBA1C	Heart issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	smoker
Id1	47.41	7.47	No	No	No	No major surgery	yes
Id2	30.36	5.77	No	No	No	No major surgery	yes
Id3	34.485	11.87	yes	No	No	2	yes
Id4	38.095	6.05	No	No	No	No major surgery	yes
Id5	35.53	5.45	No	No	No	No major surgery	yes
Id6	32.8	6.59	No	No	No	No major surgery	yes
Id7	36.4	6.07	No	No	No	No major surgery	yes
Id8	36.96	7.93	No	No	No	3	yes
Id9	41.14	9.58	yes	No	Yes	1	yes
Id10	38.06	10.79	No	No	No	No major surgery	yes
Id11	37.7	5.96	yes	No	No	2	yes
Id12	42.13	11.9	No	No	No	No major surgery	yes
Id13	40.92	8.41	No	No	No	No major surgery	yes
Id14	40.565	7.02	No	No	No	No major surgery	yes
Id15	36.385	7.59	yes	No	No	2	yes
Id16	39.9	11.32	No	No	No	No major surgery	yes
Id17	33.8	7.67	No	No	No	3	yes
Id18	36.765	7.29	yes	No	Yes	1	yes
Id19	36.955	4.72	yes	No	No	1	yes
Id20	42.9	11.41	No	No	No	No major surgery	yes
Id21	36.3	11.5	yes	No	No	2	yes
Id22	32.2	6.22	yes	No	No	2	yes
Id23	31.3	11.38	No	No	No	3	yes
Id24	41.8	7.89	yes	No	No	2	yes
Id25	35.09	4.38	yes	No	No	2	yes
Id26	33.88	7.01	No	No	No	3	yes

- Checked the Medical Examination sheet for duplicate values. No duplicate values were found.

Customer	BMI	HBA1	Heart Issue	Any Transplan	Cancer histo	NumberOfMajorSurgents	smoke
Id1	47.41	7.47	No	No	No		
Id2	30.36	5.77	No	No	No		
Id3	34.485	11.87	yes	No	No		
Id4	38.095	6.05	No	No	No		
Id5	35.53	5.45	No	No	No		
Id6	32.8	6.59	No	No	No		
Id7	36.4	6.07	No	No	No		
Id8	36.96	7.93	No	No	No		
Id9	41.14	9.58	yes	No	Yes		
Id10	38.06	10.79	No	No	No		
Id11	37.7	5.96	yes	No	No		
Id12	42.13	11.9	No	No	No		
Id13	40.92	8.41	No	No	No		
Id14	40.565	7.02	No	No	No		
Id15	36.385	7.59	yes	No	No		
Id16	39.9	11.32	No	No	No		
Id17	33.8	7.67	No	No	No		
Id18	36.765	7.29	yes	No	Yes		
Id19	36.955	4.72	yes	No	No		
Id20	42.9	11.41	No	No	No		
Id21	36.3	11.5	yes	No	No		
Id22	32.2	6.22	yes	No	No		
Id23	31.3	11.38	No	No	No		
Id24	41.8	7.89	yes	No	No		
Id25	35.09	4.38	yes	No	No		
Id26	33.88	7.01	No	No	No		
Id27						2	yes
						3	yes

- Filtered all columns and checked the values. In the Smoking column, I found some question marks (?) along with Yes and No values. I identified the mode as “No” and used Find and Replace to replace the question marks with “No”.

Customer	BMI	HBA1	Heart Issue	Any Transplan	Cancer histo	NumberOfMajorSurgents	smoke
Id1	47.41	7.47	No	No	No		yes
Id2	30.36	5.77	No	No	No		yes
Id3	34.485	11.87	yes	No	No		yes
Id4	38.095	6.05	No	No	No		yes
Id5	35.53	5.45	No	No	No		yes
Id6	32.8	6.59	No	No	No		yes
Id7	36.4	6.07	No	No	No		yes
Id8	36.96	7.93	No	No	No		yes
Id9	41.14	9.58	yes	No	Yes		yes
Id10	38.06	10.79	No	No	No		yes
Id11	37.7	5.96	yes	No	No		yes
Id12	42.13	11.9	No	No	No		yes
Id13	40.92	8.41	No	No	No		yes
Id14	40.565	7.02	No	No	No		yes
Id15	36.385	7.59	yes	No	No		yes
Id16	39.9	11.32	No	No	No		yes
Id17	33.8	7.67	No	No	No		yes
Id18	36.765	7.29	yes	No	Yes		yes
Id19	36.955	4.72	yes	No	No		yes
Id20	42.9	11.41	No	No	No		yes
Id21	36.3	11.5	yes	No	No		yes
Id22	32.2	6.22	yes	No	No		yes
Id23	31.3	11.38	No	No	No		yes
Id24	41.8	7.89	yes	No	No		yes
Id25	35.09	4.38	yes	No	No		yes
Id26	33.88	7.01	No	No	No		yes
Id27						2	yes
						3	yes

- In the “Number of Major Surgeries” column, the values were in both numerical and text formats. I converted the “No Major Surgery” text value to 0 and kept all other values in numerical format.

Find and Replace dialog box details:

- Find what: yed
- Replace with: Yes
- Within: Sheet
- Search: By Rows
- Look in: Formulas
- Match case: ☐
- Match entire cell contents: ☒

- Checked the Heart Issue and Smoking columns and corrected the inconsistent data.

Weight Status column data:

Customer ID	BMI	HBA1C	Heart Issues	Any Transplants	Cancer history	NumberOfMajorSurgeries	Smoker	Weight Status
Id1	47.41	7.47	No	No	No	0	Yes	Obesity
Id2	30.36	5.77	No	No	No	0	Yes	Obesity
Id3	34.485	11.87	Yes	No	No	2	Yes	Obesity
Id4	38.095	6.05	No	No	No	0	Yes	Obesity
Id5	35.53	5.45	No	No	No	0	Yes	Obesity
Id6	32.8	6.59	No	No	No	0	Yes	Obesity
Id7	36.4	6.07	No	No	No	0	Yes	Obesity
Id8	36.96	7.93	No	No	No	3	Yes	Obesity
Id9	41.14	9.58	Yes	No	Yes	1	Yes	Obesity
Id10	38.06	10.79	No	No	No	0	Yes	Obesity
Id11	37.7	5.96	Yes	No	No	2	Yes	Obesity
Id12	42.13	11.9	No	No	No	0	Yes	Obesity
Id13	40.92	8.41	No	No	No	0	Yes	Obesity
Id14	40.565	7.02	No	No	No	0	Yes	Obesity
Id15	36.385	7.59	Yes	No	No	2	Yes	Obesity
Id16	39.9	11.32	No	No	No	0	Yes	Obesity
Id17	33.8	7.67	No	No	No	3	Yes	Obesity
Id18	36.765	7.29	Yes	No	Yes	1	Yes	Obesity
Id19	36.955	4.72	Yes	No	No	1	Yes	Obesity
Id20	42.9	11.41	No	No	No	0	Yes	Obesity
Id21	36.3	11.5	Yes	No	No	2	Yes	Obesity
Id22	32.2	6.22	Yes	No	No	2	Yes	Obesity
Id23	31.3	11.38	No	No	No	3	Yes	Obesity
Id24	41.8	7.89	Yes	No	No	2	Yes	Obesity
Id25	35.09	4.38	Yes	No	No	2	Yes	Obesity
Id26	33.88	7.01	No	No	No	3	Yes	Obesity

- Created a new column called “Weight Status” based on BMI. Used IF conditions to classify the values as Underweight, Normal Weight, Overweight, or Obesity based on the BMI ranges.

Excel interface showing a data table with columns: Customer ID, year, month, date, children, charges, Hospital tier, City tier, State ID. The data is sorted by Customer ID, but some values are question marks (?).

Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID
?	2004	Nov		6	0	1137.01 tier - 3	tier - 1	R1013
?	1999	Jun		9	1	2775.19 tier - 2	tier - 1	R1012
?	1985	Dec		20	2	6203.9 tier - 1	tier - 2	R1012
?	2000	Oct		13	0	3585.58 tier - 1	tier - 2	R1011
?	1992	Oct		6	0	36837.47 tier - 1	tier - 2	R1011
?	1991	Nov		22	2	38711 tier - 1	tier - 3	R1011
Id1	1968	Oct		12	0	63770.43 tier - 1	tier - 3	R1013
Id10	1978	Dec		29	0	48885.14 tier - 1	tier - 2	R1013
Id100	1977	Jun		27	2	40284.38 tier - 1	tier - 3	R1012
Id1000	1989	Dec		17	3	11250.43 tier - 3	tier - 2	R1026
Id1001	1969	Dec		30	2	11244.38 tier - 3	tier - 1	R1016
Id1002	1976	Jun		28	2	11217.35 tier - 3	tier - 2	R1025
Id1003	1970	Jun		14	2	11187.66 tier - 3	tier - 2	R1012
Id1004	1972	Sep		3	0	11186.2 tier - 3	tier - 2	R1021
Id1005	1966	Aug		6	0	11165.42 tier - 3	tier - 1	R1016
Id1006	1969	Jun		25	2	11163.57 tier - 3	tier - 2	R1011
Id1007	1969	Nov		30	2	11150.78 tier - 3	tier - 2	R1011
Id1008	1980	Aug		20	2	11103.33 tier - 3	tier - 1	R1021
Id1009	1966	Jul		5	0	11093.62 tier - 3	tier - 1	R1013
Id101	1981	Oct		4	1	40273.65 tier - 1	tier - 3	R1013
Id1010	1966	Sep		9	0	11090.72 tier - 3	tier - 1	R1013
Id1011	1972	Oct		7	3	11085.59 tier - 3	tier - 2	R1012
Id1012	1967	Sep		4	0	11082.58 tier - 3	tier - 2	R1012
Id1013	1966	Nov		20	0	11073.18 tier - 3	tier - 3	R1011
Id1014	1966	Nov		7	0	11070.54 tier - 3	tier - 3	R1011
Id1015	1971	Nov		9	0	11068.77 tier - 3	tier - 2	R1012

- Next, I moved to the **Hospitalization Details** sheet and checked for duplicates using the filter option. I found some question mark (?) values in the ID column. First, I tried sorting the IDs from smallest to largest, but the sorting was not correct because the IDs were treated as text. For example, the order was coming as 1, 10, 11 instead of the proper numerical order. Since I could not delete the original data, I checked the question mark values to find the appropriate way to handle them.

Excel interface showing the same data table as above, but with the 'Customer ID' column replaced by 'id number' and the 'State ID' column replaced by 'State ID'. The data is sorted by id number, and the question mark values are replaced with 'UNKNOWN'.

id number	year	month	date	children	charges	Hospital tier	City tier	State ID
UNKNOWN_001	2004	Nov		6	0	1137.01 tier - 3	tier - 1	R1013
UNKNOWN_002	1999	Jun		9	1	2775.19 tier - 2	tier - 1	R1012
UNKNOWN_003	1985	Dec		20	2	6203.9 tier - 1	tier - 2	R1012
UNKNOWN_004	2000	Oct		13	0	3585.58 tier - 1	tier - 2	R1011
UNKNOWN_005	1992	Oct		6	0	36837.47 tier - 1	tier - 2	R1011
UNKNOWN_006	1991	Nov		22	2	38711 tier - 1	tier - 3	R1011
Id1	1968	Oct		12	0	63770.43 tier - 1	tier - 3	R1013
Id10	1978	Dec		29	0	48885.14 tier - 1	tier - 2	R1013
Id100	1977	Jun		27	2	40284.38 tier - 1	tier - 3	R1012
Id1000	1989	Dec		17	3	11250.43 tier - 3	tier - 2	R1026
Id1001	1969	Dec		30	2	11244.38 tier - 3	tier - 1	R1016
Id1002	1976	Jun		28	2	11217.35 tier - 3	tier - 2	R1025
Id1003	1970	Jun		14	2	11187.66 tier - 3	tier - 2	R1012
Id1004	1972	Sep		3	0	11186.2 tier - 3	tier - 2	R1021
Id1005	1966	Aug		6	0	11165.42 tier - 3	tier - 1	R1016
Id1006	1969	Jun		25	2	11163.57 tier - 3	tier - 2	R1011
Id1007	1969	Nov		30	2	11150.78 tier - 3	tier - 2	R1011
Id1008	1980	Aug		20	2	11103.33 tier - 3	tier - 1	R1021
Id1009	1966	Jul		5	0	11093.62 tier - 3	tier - 1	R1013
Id101	1981	Oct		4	1	40273.65 tier - 1	tier - 3	R1013
Id1010	1966	Sep		9	0	11090.72 tier - 3	tier - 1	R1013
Id1011	1972	Oct		7	3	11085.59 tier - 3	tier - 2	R1012
Id1012	1967	Sep		4	0	11082.58 tier - 3	tier - 2	R1012
Id1013	1966	Nov		20	0	11073.18 tier - 3	tier - 3	R1011
Id1014	1966	Nov		7	0	11070.54 tier - 3	tier - 3	R1011
Id1015	1971	Nov		9	0	11068.77 tier - 3	tier - 2	R1012

- Since I could not ignore the records with question marks, I replaced them with “**UNKNOWN**” and assigned different numbers to each record, such as **UNKNOWN_001**, **UNKNOWN_002**, etc., so that each ID remained unique. Next, to sort the IDs correctly in ascending order,

I converted the text IDs into numerical values using the following formula: `=IFERROR(VALUE(SUBSTITUTE(A2,"Id","")), "")`.

Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID
Id1	1968	Oct	12	0	63770.43	tier - 1	tier - 3	R1013
Id2	1977	Jun	8	0	62592.87	tier - 2	tier - 3	R1013
Id3	1970	?	11	3	60021.4	tier - 1	tier - 1	R1012
Id4	1991	Jun	6	1	58571.07	tier - 1	tier - 3	R1024
Id5	1989	Jun	19	0	55135.4	tier - 1	tier - 2	R1012
Id6	1962	Aug	4	0	52590.83	tier - 1	tier - 3	R1011
Id7	1994	Oct	27	1	51194.56	tier - 1	tier - 3	R1011
Id8	1958	Jun	27	2	49577.66	tier - 2	tier - 2	R1013
Id9	1963	Sep	4	1	48970.25	tier - 1	tier - 2	R1013
Id10	1978	Dec	29	0	48885.14	tier - 1	tier - 2	R1013
Id11	1959	Jul	22	0	48824.45	tier - 2	tier - 1	R1011
Id12	1965	Oct	27	1	48675.52	tier - 1	tier - 2	R1013
Id13	1962	Oct	11	0	48673.56	tier - 1	tier - 2	R1013
Id14	1968	Dec	1	3	48549.18	tier - 1	tier - 3	R1016
Id15	1961	Dec	21	1	48517.56	tier - 1	tier - 3	R1024
Id16	1962	Aug	27	0	48173.36	tier - 1	tier - 3	R1011
Id17	1958	Nov	16	1	47928.03	tier - 2	tier - 3	R1011
Id18	1963	Aug	5	1	47896.79	tier - 1	tier - 3	R1024
Id19	1964	Nov	7	2	47496.49	tier - 1	tier - 3	R1012
Id20	1971	Sep	27	2	47462.89	tier - 1	tier - 2	R1013
Id21	1961	Jun	13	1	47403.88	tier - 1	tier - 3	R1011
Id22	1959	Oct	4	2	47305.31	tier - 2	tier - 1	R1011
Id23	1958	Sep	3	2	47291.06	tier - 2	tier - 1	R1011
Id24	1970	Dec	28	2	47269.85	tier - 1	tier - 2	R1013
Id25	1959	Jul	14	0	47055.53	tier - 2	tier - 1	R1013
Id26	1958	Sep	15	0	46889.26	tier - 2	tier - 3	R1013

- Next, I sorted the table using the newly created ID number column and arranged the IDs in ascending order. After that, since the additional column was no longer needed, I deleted it.

Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID
Id1	1968	Oct	12	0	63770.43	tier - 1	tier - 3	R1013
Id2	1977	Jun	8	0	62592.87	tier - 2	tier - 3	R1013
Id3	1970	Sep	11	3	60021.4	tier - 1	tier - 1	R1012
Id4	1991	Jun	6	1	58571.07	tier - 1	tier - 3	R1024
Id5	1989	Jun	19	0	55135.4	tier - 1	tier - 2	R1012
Id6	1962	Aug	4	0	52590.83	tier - 1	tier - 3	R1011
Id7	1994	Oct	27	1	51194.56	tier - 1	tier - 3	R1011
Id8	1958	Jun	27	2	49577.66	tier - 2	tier - 2	R1013
Id9	1963	Sep	4	1	48970.25	tier - 1	tier - 2	R1013
Id10	1978	Dec	29	0	48885.14	tier - 1	tier - 2	R1013
Id11	1959	Jul	22	0	48824.45	tier - 2	tier - 1	R1011
Id12	1965	Oct	27	1	48675.52	tier - 1	tier - 2	R1013
Id13	1962	Oct	11	0	48673.56	tier - 1	tier - 2	R1013
Id14	1968	Dec	1	3	48549.18	tier - 1	tier - 3	R1016
Id15	1961	Dec	21	1	48517.56	tier - 1	tier - 3	R1024
Id16	1962	Aug	27	0	48173.36	tier - 1	tier - 3	R1011
Id17	1958	Nov	16	1	47928.03	tier - 2	tier - 3	R1011
Id18	1963	Aug	5	1	47896.79	tier - 1	tier - 3	R1024
Id19	1964	Nov	7	2	47496.49	tier - 1	tier - 3	R1012
Id20	1971	Sep	27	2	47462.89	tier - 1	tier - 2	R1013
Id21	1961	Jun	13	1	47403.88	tier - 1	tier - 3	R1011
Id22	1959	Oct	4	2	47305.31	tier - 2	tier - 1	R1011
Id23	1958	Sep	3	2	47291.06	tier - 2	tier - 1	R1011
Id24	1970	Dec	28	2	47269.85	tier - 1	tier - 2	R1013
Id25	1959	Jul	14	0	47055.53	tier - 2	tier - 1	R1013
Id26	1958	Sep	15	0	46889.26	tier - 2	tier - 3	R1013

- Next, I imputed the missing **Month** values as **September** and filled the missing **Year** values using the average of the available years. I then rounded the average to the nearest whole number and used it to fill the missing years.

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Ready Accessibility: Good to go

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	Customer	year	month	date	children	charges	Hospital tier	City tier	State ID										
2	Id1	1968	Oct	12	0	63770.43	tier - 1	tier - 3	R1013										
3	Id2	1977	Jun	8	0	62592.87	tier - 2	tier - 3	R1013										
4	Id3	1970	Sep	11	3	60021.4	tier - 1	tier - 1	R1012										
5	Id4	1991	Jun	6	1	58571.07	tier - 1	tier - 3	R1024										
6	Id5	1989	Jun	19	0	55135.4	tier - 1	tier - 2	R1012										
7	Id6	1962	Aug	4	0	52590.83	tier - 1	tier - 3	R1011										
8	Id7	1994	Oct	27	1	51194.56	tier - 1	tier - 3	R1011										
9	Id8	1958	Jun	27	2	49577.66	tier - 2	tier - 2	R1013										
10	Id9	1963	Sep	4	1	48970.25	tier - 1	tier - 2	R1013										
11	Id10	1978	Dec	29	0	48885.14	tier - 1	tier - 2	R1013										
12	Id11	1959	Jul	22	0	48824.45	tier - 2	tier - 1	R1011										
13	Id12	1965	Oct	27	1	48675.52	tier - 1	tier - 2	R1013										
14	Id13	1962	Oct	11	0	48673.56	tier - 1	tier - 2	R1013										
15	Id14	1968	Dec	1	3	48549.18	tier - 1	tier - 3	R1016										
16	Id15	1961	Dec	21	1	48517.56	tier - 1	tier - 3	R1024										
17	Id16	1962	Aug	27	0	48173.36	tier - 1	tier - 3	R1011										
18	Id17	1958	Nov	16	1	47928.03	tier - 2	tier - 3	R1011										
19	Id18	1963	Aug	5	1	47896.79	tier - 1	tier - 3	R1024										
20	Id19	1964	Nov	7	2	47496.49	tier - 1	tier - 3	R1012										
21	Id20	1971	Sep	27	2	47462.89	tier - 1	tier - 2	R1013										
22	Id21	1961	Jun	13	1	47403.88	tier - 1	tier - 3	R1011										
23	Id22	1959	Oct	4	2	47305.31	tier - 2	tier - 1	R1011										
24	Id23	1958	Sep	3	2	47291.06	tier - 2	tier - 1	R1011										
25	Id24	1970	Dec	28	2	47269.85	tier - 1	tier - 2	R1013										
26	Id25	1959	Jul	14	0	47055.53	tier - 2	tier - 1	R1013										
27	Id26	1958	Sep	15	0	46889.26	tier - 2	tier - 3	R1013										

Customer Names Medical Examinations Hospitalisation Details

average value of year column round the average value of year 1982.842 1983

hospital tier mode value tier - 2

city tier mode value tier - 2

- Next, I filled the missing values in the **Hospital Tier** and **City Tier** columns using the **most frequently occurring values (mode)**. Then, I filled the missing values in the **State ID** column with **“Unknown”**.

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Ready Accessibility: Good to go

Average: 40.158344 Count: 2343 Sum: 94091

100%

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID	Date Of Birth	Age								
2	Id1	1968	Oct	12	0	₹ 63,770.43	tier - 1	tier - 3	R1013	12-Oct-1968	55								
3	Id2	1977	Jun	8	0	₹ 62,592.87	tier - 2	tier - 3	R1013	8-Jun-1977	46								
4	Id3	1970	Sep	11	3	₹ 60,021.40	tier - 1	tier - 1	R1012	11-Sep-1970	53								
5	Id4	1991	Jun	6	1	₹ 58,571.07	tier - 1	tier - 3	R1024	6-Jun-1991	32								
6	Id5	1989	Jun	19	0	₹ 55,135.40	tier - 1	tier - 2	R1012	19-Jun-1989	34								
7	Id6	1962	Aug	4	0	₹ 52,590.83	tier - 1	tier - 3	R1011	4-Aug-1962	61								
8	Id7	1994	Oct	27	1	₹ 51,194.56	tier - 1	tier - 3	R1011	27-Oct-1994	29								
9	Id8	1958	Jun	27	2	₹ 49,577.66	tier - 2	tier - 2	R1013	27-Jun-1958	65								
10	Id9	1963	Sep	4	1	₹ 48,970.25	tier - 1	tier - 2	R1013	4-Sep-1963	60								
11	Id10	1978	Dec	29	0	₹ 48,885.14	tier - 1	tier - 2	R1013	29-Dec-1978	45								
12	Id11	1959	Jul	22	0	₹ 48,824.45	tier - 2	tier - 1	R1011	22-Jul-1959	64								
13	Id12	1965	Oct	27	1	₹ 48,675.52	tier - 1	tier - 2	R1013	27-Oct-1965	58								
14	Id13	1962	Oct	11	0	₹ 48,673.56	tier - 1	tier - 2	R1013	11-Oct-1962	61								
15	Id14	1968	Dec	1	3	₹ 48,549.18	tier - 1	tier - 3	R1016	1-Dec-1968	55								
16	Id15	1961	Dec	21	1	₹ 48,517.56	tier - 1	tier - 3	R1024	21-Dec-1961	62								
17	Id16	1962	Aug	27	0	₹ 48,173.36	tier - 1	tier - 3	R1011	27-Aug-1962	61								
18	Id17	1958	Nov	16	1	₹ 47,928.03	tier - 2	tier - 3	R1011	16-Nov-1958	65								
19	Id18	1963	Aug	5	1	₹ 47,896.79	tier - 1	tier - 3	R1024	5-Aug-1963	60								
20	Id19	1964	Nov	7	2	₹ 47,496.49	tier - 1	tier - 3	R1012	7-Nov-1964	59								
21	Id20	1971	Sep	27	2	₹ 47,462.89	tier - 1	tier - 2	R1013	27-Sep-1971	52								
22	Id21	1961	Jun	13	1	₹ 47,403.88	tier - 1	tier - 3	R1011	13-Jun-1961	62								
23	Id22	1959	Oct	4	2	₹ 47,305.31	tier - 2	tier - 1	R1011	4-Oct-1959	64								
24	Id23	1958	Sep	3	2	₹ 47,291.06	tier - 2	tier - 1	R1011	3-Sep-1958	65								
25	Id24	1970	Dec	28	2	₹ 47,269.85	tier - 1	tier - 2	R1013	28-Dec-1970	53								
26	Id25	1959	Jul	14	0	₹ 47,055.53	tier - 2	tier - 1	R1013	14-Jul-1959	64								
27	Id26	1958	Sep	15	0	₹ 46,889.26	tier - 2	tier - 3	R1013	15-Sep-1958	65								

Customer Names Medical Examinations Hospitalisation Details

average value of round the average

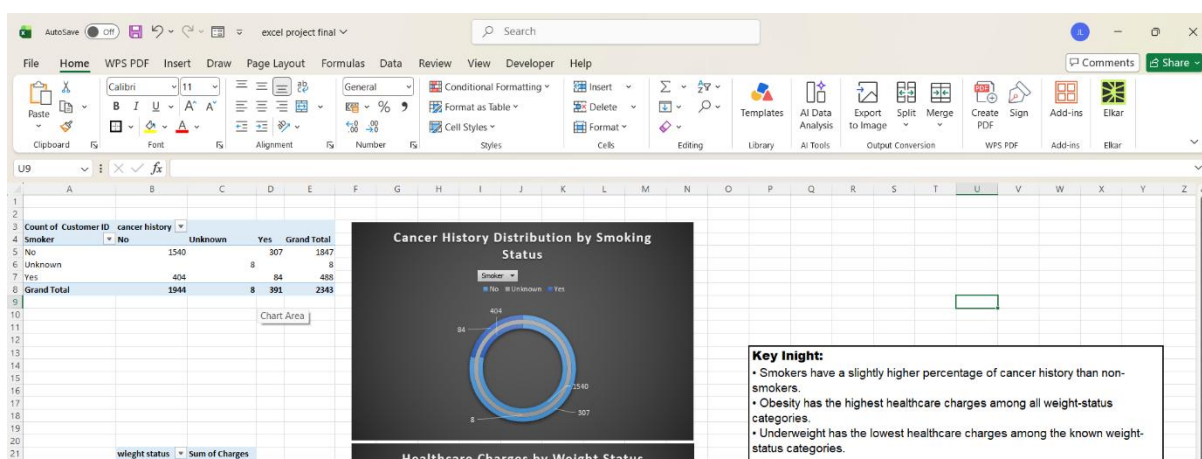
hospital tier m

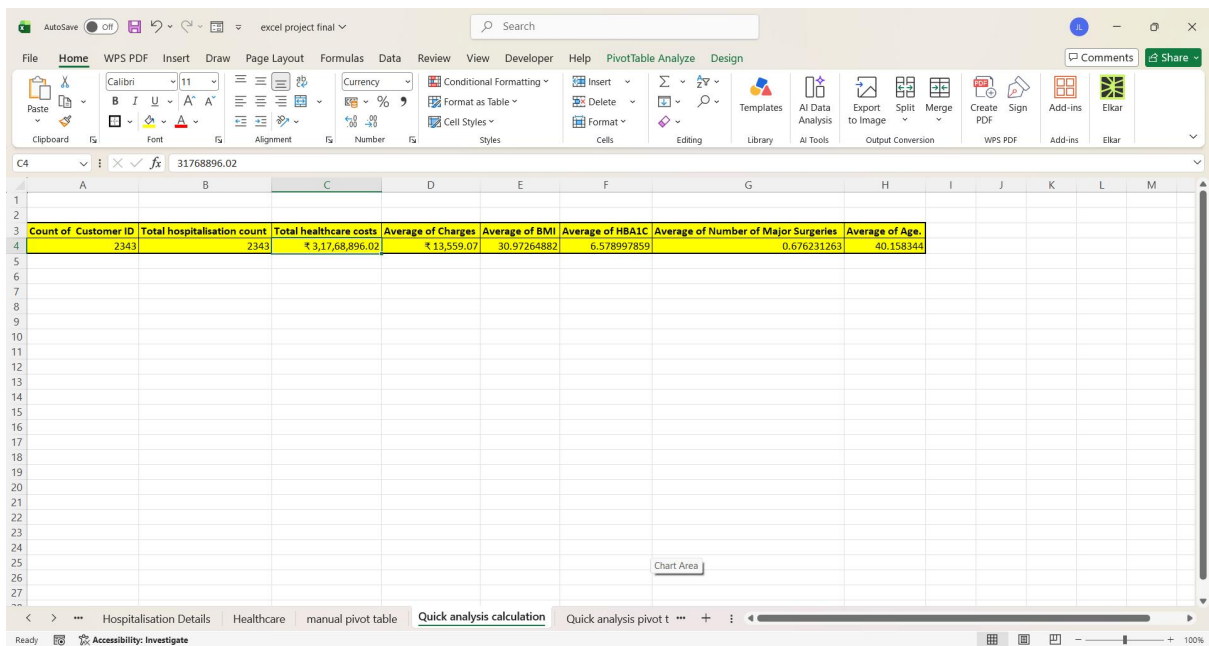
city tier mo

- Next, I merged the **Year**, **Month**, and **Date** columns into a single **Date of Birth** column. Then, I calculated the **age of each customer** based on the Date of Birth as of **8th June 2023**. Finally, I changed the **currency format** in the **Charge** column.

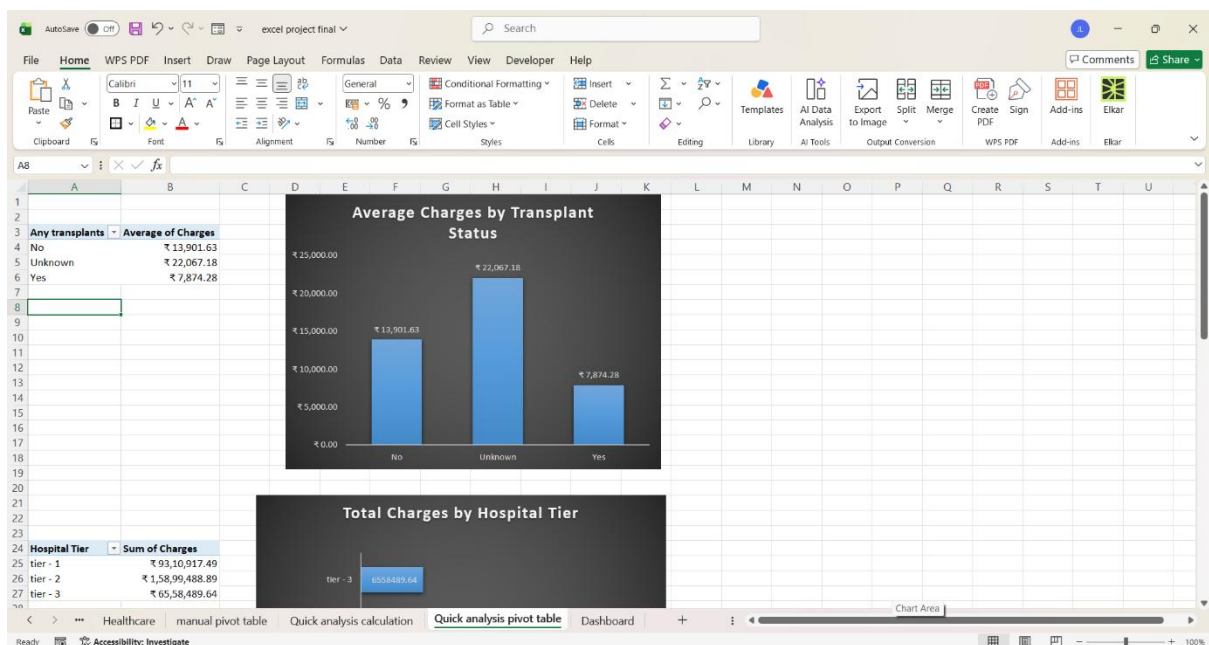
	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	BMI	HBA1c	Heart Issue	Any Transplant	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Diabetes Status	Date of Birth	Charges	Hospital Tier	City Tier	State	Age		
1	47.41	7.47	No	No	No		0	Yes	Obesity	Diabetes	12-Oct-1968	₹ 63,770.43	tier - 1	tier - 3	R1013	55	
2	30.36	5.77	No	No	No		0	Yes	Obesity	Diabetes	8-Jun-1977	₹ 62,592.87	tier - 2	tier - 3	R1013	46	
3	34.485	11.87	Yes	No	No		2	Yes	Obesity	Diabetes	11-Sep-1970	₹ 60,021.40	tier - 1	tier - 1	R1012	53	
4	38.095	6.05	No	No	No		0	Yes	Obesity	Prediabetes	6-Jun-1991	₹ 58,571.07	tier - 1	tier - 3	R1024	32	
5	35.53	5.45	No	No	No		0	Yes	Obesity	Normal	19-Jun-1989	₹ 55,135.40	tier - 1	tier - 2	R1012	34	
6	32.8	6.59	No	No	No		0	Yes	Obesity	Diabetes	4-Aug-1962	₹ 52,590.83	tier - 1	tier - 3	R1011	61	
7	36.4	6.07	No	No	No		0	Yes	Obesity	Prediabetes	27-Oct-1994	₹ 51,194.56	tier - 1	tier - 3	R1011	29	
8	36.96	7.93	No	No	No		3	Yes	Obesity	Diabetes	27-Jun-1958	₹ 49,577.66	tier - 2	tier - 2	R1013	65	
9	41.14	9.58	Yes	No	Yes		1	Yes	Obesity	Diabetes	4-Sep-1963	₹ 48,970.25	tier - 1	tier - 2	R1013	60	
10	38.06	10.79	No	No	No		0	Yes	Obesity	Diabetes	29-Dec-1978	₹ 48,885.14	tier - 1	tier - 2	R1013	45	
11	37.7	5.96	Yes	No	No		2	Yes	Obesity	Prediabetes	22-Jul-1959	₹ 48,824.45	tier - 2	tier - 1	R1011	64	
12	42.13	11.9	No	No	No		0	Yes	Obesity	Diabetes	27-Oct-1965	₹ 48,675.52	tier - 1	tier - 2	R1013	58	
13	40.92	8.43	No	No	No		0	Yes	Obesity	Diabetes	11-Oct-1962	₹ 48,673.56	tier - 1	tier - 2	R1013	61	
14	40.565	7.02	No	No	No		0	Yes	Obesity	Diabetes	1-Dec-1968	₹ 48,549.18	tier - 1	tier - 3	R1016	55	
15	36.385	7.59	Yes	No	No		2	Yes	Obesity	Diabetes	21-Dec-1961	₹ 48,517.56	tier - 1	tier - 3	R1024	62	
16	39.9	11.32	No	No	No		0	Yes	Obesity	Diabetes	27-Aug-1962	₹ 48,173.36	tier - 1	tier - 3	R1011	61	
17	33.8	7.67	No	No	No		3	Yes	Obesity	Diabetes	16-Nov-1958	₹ 47,928.03	tier - 2	tier - 3	R1011	65	
18	36.765	7.29	Yes	No	Yes		1	Yes	Obesity	Diabetes	5-Aug-1963	₹ 47,896.79	tier - 1	tier - 3	R1024	60	
19	36.955	4.72	Yes	No	No		1	Yes	Obesity	Normal	7-Nov-1964	₹ 47,496.49	tier - 1	tier - 3	R1012	59	
20	42.9	11.41	No	No	No		0	Yes	Obesity	Diabetes	27-Sep-1971	₹ 47,462.89	tier - 1	tier - 2	R1013	52	
21	36.3	11.5	Yes	No	No		2	Yes	Obesity	Diabetes	13-Jun-1961	₹ 47,403.88	tier - 1	tier - 3	R1011	62	
22	32.2	6.22	Yes	No	No		2	Yes	Obesity	Prediabetes	4-Oct-1959	₹ 47,305.31	tier - 2	tier - 1	R1011	64	
23	31.3	11.38	No	No	No		3	Yes	Obesity	Diabetes	3-Sep-1958	₹ 47,291.06	tier - 2	tier - 1	R1011	65	
24	41.8	7.89	Yes	No	No		2	Yes	Obesity	Diabetes	28-Dec-1970	₹ 47,269.85	tier - 1	tier - 2	R1013	53	
25	35.09	4.38	Yes	No	No		2	Yes	Obesity	Normal	14-Jul-1959	₹ 47,055.53	tier - 2	tier - 1	R1013	64	
26	33.88	7.01	No	No	No		2	Yes	Obesity	Diabetes	15-Sep-1958	₹ 46,889.76	tier - 2	tier - 3	R1013	65	

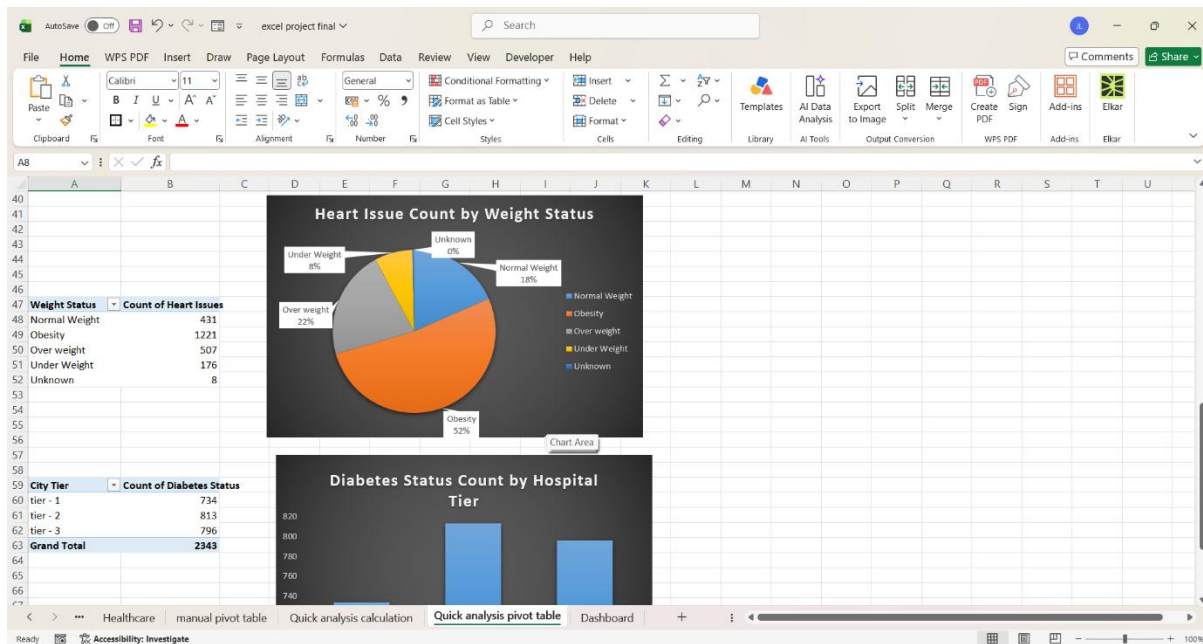
- Next, I created a new sheet and added all the required columns mentioned in the instructions. I used **XLOOKUP** to retrieve and add the required information from the cleaned data. For the **Unknown IDs**, the actual information was not available in the dataset. So, I replaced the missing values in **categorical columns** with “**Unknown.**” For **numerical columns**, I calculated the **average value** and used it to fill the missing values for the Unknown IDs..



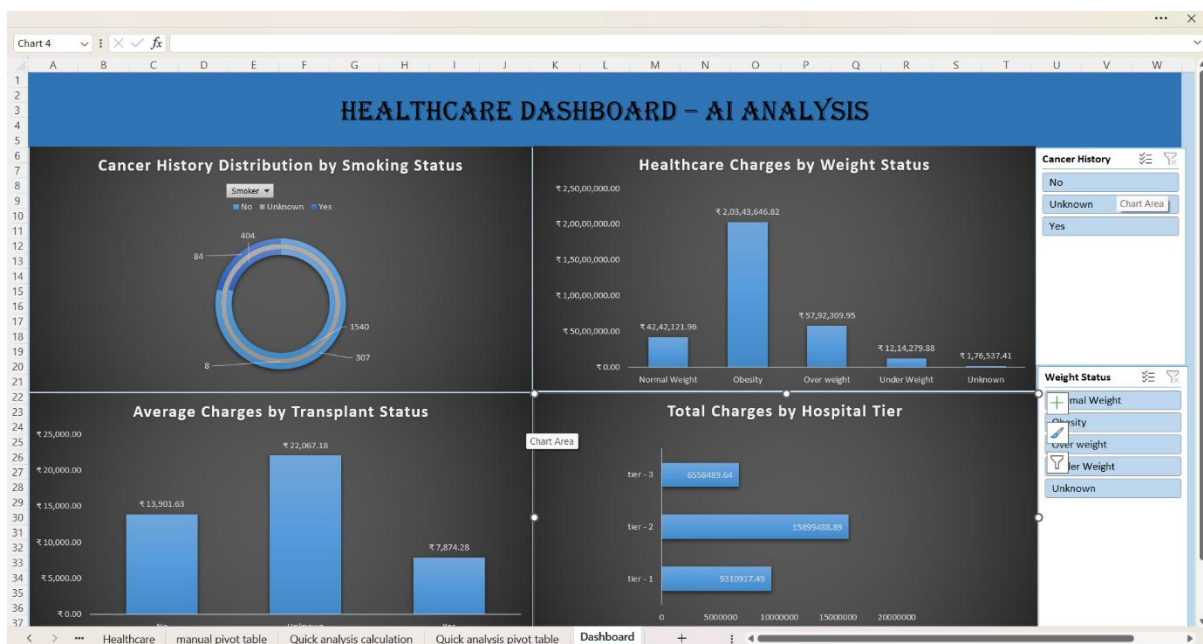


- Next, I created a new sheet called **Quick Analysis Calculation**. Using **Quick Analysis**, I performed the calculations mentioned in the instructions, including **Total Customer Count**, **Total Hospitalization Count**, **Total Healthcare Cost**, **Average Healthcare Cost**, and **Medical Metrics**.





- Next, I created **Quick Analysis Pivot Tables** for the required analysis. I created PivotTables for **Average Charges by Transplant Status**, **Total Charges by Hospital Tier**, **Heart issue Count by Weight Status**, and **Diabetes Status Count by Hospital Tier**. I then created charts for these PivotTables using **AI-assisted chart recommendations**



- Finally, I created a new sheet named **Dashboard** and built my **Healthcare Dashboard with AI Analysis**. I added all the required charts to the dashboard. I also added slicers for **Weight Status** and **Diabetes Status**. From my point of view, the **Cancer history distribution by Smoking Status** needed a little more clarification, so I also added a **Cancer history slicer** to

make the analysis more interactive and easier to understand. At the end of the process, I created my **first interactive dashboard**.